Computer Networks

Module-3 :-

$$S = G * e^{-2G}$$

$S \approx$ Throughput

G = Generated frames

$$S_{max} = 18.4\%$$

when $G = 1/2$

Vulnerable time = 2 . Tfr .)

Data Link LAYER

1. Framing
2. Address
3. Flow control
4. Access control
5. Error Handling

Data pckts encapsulated into frames

Sol Given :

ALOHA channel capacity $= 56 kbps \approx 56000 bps$

Frame size $=$ 1000 bit

Frame generation time $=$ 100 sec

To find : $N_{max} = ?$ (max. no. of. stations).

$$\boxed{N * G = \frac{1}{2e}} \qquad e \approx 2.178$$

$$\boxed{\rightarrow \text{Frame transmission time} = \frac{\text{Frame size}}{\text{Channel capacity}}}$$

$$= \frac{1000}{56000} = 0.017$$

$$= 0.0178 \text{ seconds.}$$

$\rightarrow$ Vulnerable time $= 2 \times T_{fr}$

$$= 2 \times 0.0178$$

$$= 0.0356$$

$\rightarrow$ To find G : $G = (\text{no. of. frames/sec}) * \text{frame transmission time.}$

$$G = \left(\frac{1}{100}\right)(0.0178)$$

$$= 0.000178$$

Post-MIDS

HTTP: ⟹ HTTPS

* Hyper-text-transfer-protocol

→ Information is sent as clear text
→ Vulnerable to Hackers
→ Discorded and introduced HTTPS

HTTPS

→ Secured version of HTTPS
→ Highly encrypted.
→ Information will be unreadable when hacked / tampered.

Subnet Masks:

* Identifies the network portion & Host portion of a IP address.

IP address: 8 Bit Octet Table

Class A (0-126)
Class B (128-191)
Class C (192-223)

Network 255	Host 0	Host 0	Host 0	→ Class A
Network 255	Network 255	Host 0	Host 0	→ Class B
Network 255	Network 255	Network 255	Host 0	→ Class C

No. of Hosts:

Class A: 254 × 254 × 254 = 16 M Computers

Class B: 254 × 254 computers

Class C: 254 Hosts / Computers

P Eg: IP: 192.168.1.0

255.255.255.0

Default Subnet

Step-1 : find Binary values

Octet table :

* Put '1'

wherever the sum of them tends to give the IP address

128+64 = 192

192.168.1.0

128	64	32	16	8	4	2	1

192	1	1	0	0	0	0	0	0
168	1	0	1	0	1	0	0	0
1	0	0	0	0	0	0	0	1
0	0	0	0	0	0	0	0	0

Binary : 192.168.1.0 ⟹

11000000.10101000.00000001.00000000

24.12.24

1225 H

OSI ; TCP/IP.

OSI : All people seem to Need Data Processing.

TCP/IP

- A — Application layer
- P — Presentation layer
- S — Session layer
- T — Transport layer
- N — Network
- D — Data-link
- P — Physical

} Additional in OSI

* OSI is advanced form of TCP/IP.

(P) Check Sum : 7, 11, 12, 0, 6

Sol

$(7)_{10} \Rightarrow (111)_2$

$(11)_{10} \Rightarrow (1011)_2$

$(12)_{10} \Rightarrow (1100)_2$

$(0)_{10} \Rightarrow (0000)_2$

$(6)_{10} \Rightarrow (110)_2$

<u>Sender</u> :

* Generator divides the msg to m-bits

* Gen. also adds +m bit

<u>Receiver</u>:

* The receiver creates new checksum.

* if (new checksum == 0) {
    msg.accept();
  }
  else {
    msg.reject();
  } # Shows Error

Eg:- 7, 11, 12, 0, 6

decimal (10)

7 $(111)_2$

11 $(1011)_2$

12 $(1100)_2$

0 $(0000)_2$

6 $(110)_2$

Step-1:

$$
\begin{array}{r}
\ 111 \\
1011 \\
1100 \\
0000 \\
+\ \ 110 \\
\hline
\underbrace{10}\,0100
\end{array}
$$

Must Consider only upto 4 Bits.

So, add the rest

$$
\begin{array}{r}
0100 \\
+\ \ 10 \\
\hline
0110
\end{array}
$$

Now, take 1's complement:

0110

$(1001) \Rightarrow (9)$

Value to be sent with the message.

$(7, 11, 12, 0, 6, 9) \Rightarrow$ Message sent to the receiver.

At Receiver's End :-

7 $(111)_2$

11 $(1011)_2$

12 $(1100)_2$

0 $(0000)_2$

6 $(110)_2$

9 (1001)

$$
\begin{array}{r}
111 \\
1011 \\
1100 \\
0000 \\
+ \quad 110 \\
\hline
101101 \\
\hline
\end{array}
$$

101101

Consider 4 Bits
Rest of them add.

$$
\begin{array}{r}
1101 \\
+ \quad 10 \\
\hline
1111 \\
\hline
\end{array}
$$

Find 1s complement

$(1111) \Rightarrow (0000)_2$

new checksum $= (0)_{10}$

msg.accept ( );

∴ Message will be accepted.